

Foreword

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where:

x the first digit:

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y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present document establishes the minimum RF requirements for NR User Equipment (UE) operating on frequency Range 1.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document. References are either specific (identified by date of publication, edition number, version number, etc.) or nonspecific.

For a specific reference, subsequent revisions do not apply.

For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in the same Release as the present document.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone".
- [3] 3GPP TS 38.101-3: "NR; User Equipment (UE) radio transmission and reception; Part 3: Range 1 and Range 2 Interworking operation with other radios".
- [4] 3GPP TS 38.521-1: "NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Range 1 Standalone".
- [5] Recommendation ITU-R M.1545: "Measurement uncertainty as it applies to test limits for the terrestrial component of International Mobile Telecommunications-2000".
- [6] 3GPP TS 38.211: "NR; Physical channels and modulation".
- [7] 3GPP TS 38.331: "Radio Resource Control (RRC) protocol specification".
- [8] 3GPP TS 38.213: "NR; Physical layer procedures for control".
- [9] ITU-R Recommendation SM.329-10, "Unwanted emissions in the spurious domain".
- [10] 3GPP TS 38.214: "NR; Physical layer procedures for data".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

Aggregated Channel Bandwidth: The RF bandwidth in which a UE transmits and receives multiple contiguously aggregated carriers.

Carrier aggregation: Aggregation of two or more component carriers in order to support wider transmission bandwidths.

Carrier aggregation band: A set of one or more operating bands across which multiple carriers are aggregated with a specific set of technical requirements.

Carrier aggregation bandwidth class: A class defined by the aggregated transmission bandwidth configuration and maximum number of component carriers supported by a UE.

Carrier aggregation configuration: A combination of CA operating band(s) and CA bandwidth class(es) supported by a UE.

Contiguous carriers: A set of two or more carriers configured in a spectrum block where there are no RF requirements based on co-existence for un-coordinated operation within the spectrum block.

Contiguous resource allocation: A resource allocation of consecutive resource blocks within one carrier or across contiguously aggregated carriers. The gap between contiguously aggregated carriers due to the nominal channel spacing is allowed.

Contiguous spectrum: Spectrum consisting of a contiguous block of spectrum with no sub-block gaps.

Inter-band carrier aggregation: Carrier aggregation of component carriers in different operating bands.

NOTE: Carriers aggregated in each band can be contiguous or non-contiguous.

Intra-band contiguous carrier aggregation: Contiguous carriers aggregated in the same operating band.

Intra-band non-contiguous carrier aggregation: Non-contiguous carriers aggregated in the same operating band.

Sub-block: This is one contiguous allocated block of spectrum for transmission and reception by the same UE. There may be multiple instances of sub-blocks within an RF bandwidth.

Sub-block bandwidth: The bandwidth of one sub-block.

Sub-block gap: A frequency gap between two consecutive sub-blocks within an RF bandwidth, where the RF requirements in the gap are based on co-existence for un-coordinated operation.

UE transmission bandwidth configuration: Set of resource blocks located within the UE channel bandwidth which may be used for transmitting or receiving by the UE.

Vehicular UE: A UE embedded in a vehicle, permanently connected to an embedded antenna system that radiates externally for NR operating bands.

NOTE: Vehicular UE does not refer to other UE form factors placed inside the vehicle.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

ΔF_{Global}	Granularity of the global frequency raster
ΔF_{Raster}	Band dependent channel raster granularity
Δf_{OOB}	Δ Frequency of Out Of Band emission
$\Delta F_{\text{TX-RX}}$	Δ Frequency of default TX-RX separation of the FDD <i>operating band</i>
$\Delta P_{\text{PowerClass}}$	Adjustment to maximum output power for a given power class
$\frac{F_0}{4.4}_{\text{RB}}$	The starting frequency offset between the allocated RB and the measured non-allocated RB
$\Delta R_{\text{IB},c}$	Allowed reference sensitivity relaxation due to support for inter-band CA operation, for serving cell c
$\Delta R_{\text{IB},4R}$	Reference sensitivity adjustment due to support for 4 antenna ports
Δ_{Shift}	Channel raster offset
$\frac{F_0}{4.4}_{\text{TC}}$	Allowed operating band edge transmission power relaxation
$\frac{F_0}{4.4}_{\text{TC},c}$	Allowed operating band edge transmission power relaxation for serving cell c
$\Delta T_{\text{IB},c}$	Allowed maximum configured output power relaxation due to support for inter-band CA operation and due to support for SUL operations, for serving cell c
BW_{Channel}	Channel bandwidth
$BW_{\text{Channel,block}}$	Sub-block bandwidth, expressed in MHz. $BW_{\text{Channel,block}} = F_{\text{edge,block,high}} - F_{\text{edge,block,low}}$
$BW_{\text{Channel_CA}}$	Aggregated channel bandwidth, expressed in MHz
$BW_{\text{Channel,max}}$	Maximum channel bandwidth supported among all bands in a release
BW_{GB}	$\max(BW_{\text{GB,Channel}(k)})$
$BW_{\text{GB,Channel}(k)}$	Minimum guard band defined in clause 5.3A.1 of carrier k
BW_{DL}	Channel bandwidth for DL
BW_{UL}	Channel bandwidth for UL
$BW_{\text{interferer}}$	Bandwidth of the interferer

Ceil(x)	Rounding upwards; ceil(x) is the smallest integer such that $\text{ceil}(x) \geq x$
Floor(x)	Rounding downwards; floor(x) is the greatest integer such that $\text{floor}(x) \leq x$
F_C	Center frequency of a carrier for a numerology defined by the <i>RF reference frequency</i> on the channel raster mapped to the carrier according to sub-clause 5.4.2.2
$F_{C,\text{block, high}}$	F_C of the highest transmitted/received carrier in a <i>sub-block</i>
$F_{C,\text{block, low}}$	F_C of the lowest transmitted/received carrier in a <i>sub-block</i>
$F_{C,\text{low}}$	The F_C of the lowest carrier, expressed in MHz
$F_{C,\text{high}}$	The F_C of the highest carrier, expressed in MHz
$F_{\text{DL,low}}$	The lowest frequency of the downlink <i>operating band</i>
$F_{\text{DL,high}}$	The highest frequency of the downlink <i>operating band</i>
$F_{\text{UL,low}}$	The lowest frequency of the uplink <i>operating band</i>
$F_{\text{UL,high}}$	The highest frequency of the uplink <i>operating band</i>
$F_{\text{edge,block,low}}$	The lower <i>sub-block</i> edge, where $F_{\text{edge,block,low}} = F_{C,\text{block,low}} - F_{\text{offset, low}}$.
$F_{\text{edge,block,high}}$	The upper <i>sub-block</i> edge, where $F_{\text{edge,block,high}} = F_{C,\text{block,high}} + F_{\text{offset, high}}$.
$F_{\text{edge, low}}$	The <i>lower edge</i> of <i>aggregated channel bandwidth</i> , expressed in MHz. $F_{\text{edge,low}} = F_{C,\text{low}} - F_{\text{offset,low}}$.
$F_{\text{edge, high}}$	The <i>higher edge</i> of <i>aggregated channel bandwidth</i> , expressed in MHz. $F_{\text{edge,high}} = F_{C,\text{high}} + F_{\text{offset,high}}$.
$F_{\text{Interferer (offset)}}$	Frequency offset of the interferer (between the center frequency of the interferer and the carrier frequency of the carrier measured). For intra-band contiguous CA, the $F_{\text{Interferer (offset)}}$ is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the interferer
$F_{\text{Interferer}}$	Frequency of the interferer
F_{Ioffset}	Frequency offset of the interferer (between the center frequency of the interferer and the closest edge of the carrier measured)
F_{offset}	Frequency offset from $F_{C,\text{high}}$ to the <i>higher edge</i> or $F_{C,\text{low}}$ to the <i>lower edge</i> .
$F_{\text{offset,high}}$	Frequency offset from $F_{C,\text{high}}$ to the upper <i>UE RF Bandwidth edge</i> , or from $F_{C,\text{block, high}}$ to the upper sub-block edge
$F_{\text{offset,low}}$	Frequency offset from $F_{C,\text{low}}$ to the lower <i>UE RF Bandwidth edge</i> , or from $F_{C,\text{block, low}}$ to the lower sub-block edge
F_{OOB}	The boundary between the NR out of band emission and spurious emission domains
F_{REF}	RF reference frequency
$F_{\text{REF-Offs}}$	Offset used for calculating F_{REF}
$F_{\text{REF, shift}}$	RF reference frequency for Supplementary Uplink (SUL) bands, the uplink of all FDD bands, and TDD bands
$F_{\text{uw (offset)}}$	The frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the interferer
$\text{GB}_{\text{Channel}}$	Minimum guard band defined in clause 5.3.3, expressed in kHz

L_{CRB}	Transmission bandwidth which represents the length of a contiguous resource block allocation expressed in units of resources blocks
Max()	The largest of given numbers
Min()	The smallest of given numbers
	Physical resource block number
NR_{ACLR}	NR ACLR
N_{RB}	Transmission bandwidth configuration, expressed in units of resource blocks
$N_{\text{RB_agg}}$	The number of the aggregated RBs within the fully allocated aggregated channel bandwidth for carrier 1 to j, where μ is defined in TS 38.211 [6]
$N_{\text{RB},c}$	The transmission bandwidth configuration of component carrier c, expressed in units of resource blocks for carrier j, where μ is defined in TS 38.211 [6]
$N_{\text{RB},\text{low}}$	The transmission bandwidth configurations according to Table 5.3.2-1 for the lowest assigned component carrier in section 5.3A.1
$N_{\text{RB},\text{high}}$	The transmission bandwidth configurations according to Table 5.3.2-1 for the highest assigned component carrier in section 5.3A.1
N_{REF}	NR Absolute Radio Frequency Channel Number (NR-ARFCN)
$N_{\text{REF-Offs}}$	Offset used for calculating N_{REF}
P_{CMAX}	The configured maximum UE output power
$P_{\text{CMAX}, c}$	The configured maximum UE output power for serving cell c
$P_{\text{CMAX}, f, c}$	The configured maximum UE output power for carrier f of serving cell c in each slot
P_{EMAX}	Maximum allowed UE output power signalled by higher layers
$P_{\text{EMAX}, c}$	Maximum allowed UE output power signalled by higher layers for serving cell c
$P_{\text{Interferer}}$	Modulated mean power of the interferer
$P_{\text{largest BW}}$	Power of the largest transmission bandwidth configuration of the component carriers in the bandwidth combination
$P_{\text{PowerClass}}$	The nominal UE power (i.e., no tolerance)
$P\text{-MPR}_c$	Power Management Maximum Power Reduction for serving cell c
P_{RB}	The transmitted power per allocated RB, measured in dBm
P_{UMAX}	The measured configured maximum UE output power
P_{uw}	Power of an unwanted DL signal
P_{w}	Power of a wanted DL signal
RB_{start}	The lowest RB index of transmitted resource blocks
SCS_c	SCS for the component carrier c, expressed in kHz

$SCS_{\text{largest BW}}$	SCS for the largest transmission bandwidth configuration of the component carriers in the bandwidth combination, expressed in kHz
SCS_{low}	SCS for the lowest assigned component carrier in section 5.3A.1, expressed in kHz
SCS_{high}	SCS for the highest assigned component carrier in section 5.3A.1, expressed in kHz
$T(P_{\text{CMAX}, f_c})$	Tolerance for applicable values of P_{CMAX, f_c} for configured maximum UE output power for carrier f of serving cell c
$T_{L,c}$	Absolute value of the lower tolerance for the applicable <i>operating band</i> as specified in section 6.2.1
SS_{REF}	SS block reference frequency position
$UTRA_{\text{ACLR}}$	UTRA ACLR

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

ACLR	Adjacent Channel Leakage Ratio
ACS	Adjacent Channel Selectivity
A-MPR	Additional Maximum Power Reduction
BS	Base Station
BW	Bandwidth
BWP	Bandwidth Part
CA	Carrier Aggregation
CA_{nX-nY}	Inter-band CA of component carrier(s) in one sub-block within Band nX and component carrier(s) in one sub-block within Band nY where nX and nY are the applicable NR <i>operating bands</i>
CC	Component Carriers
CP-OFDM	Cyclic Prefix-OFDM
CW	Continuous Wave
DC	Dual Connectivity
DFT-s-OFDM	Discrete Fourier Transform-spread-OFDM
DM-RS	Demodulation Reference Signal
DTX	Discontinuous TransmissionE-UTRA Evolved UTRA
EVM	Error Vector Magnitude
FR	Frequency Range
FRC	Fixed Reference Channel
GSCN	Global Synchronization Channel Number

IBB	In-band Blocking
IDFT	Inverse Discrete Fourier Transformation
ITUR	Radiocommunication Sector of the International Telecommunication Union
MBW	Measurement bandwidth
MOP	Maximum Output Power
MPR	Allowed maximum power reduction
MSD	Maximum Sensitivity Degradation
NR	New Radio
NR-ARFCN	NR Absolute Radio Frequency Channel Number
NS	Network Signalling
OCNG	OFDMA Channel Noise Generator
OOB	Out-of-band
P-MPR	Power Management Maximum Power Reduction
PRB	Physical Resource Block
QAM	Quadrature Amplitude Modulation
RE	Resource Element
REFSENS	Reference Sensitivity
RF	Radio Frequency
RMS	Root Mean Square (value)
RSRP	Reference Signal Receiving Power
Rx	Receiver
SC	Single Carrier
SCS	Subcarrier spacing
SDL	Supplementary Downlink
SEM	Spectrum Emission Mask
SNR	Signal-to-Noise Ratio
SRS	Sounding Reference Symbol
SUL	Supplementary uplink
SS	Synchronization Symbol
SUL	Supplementary uplink
TAE	Time Alignment Error
Tx	Transmitter
UL MIMO	Uplink Multiple Antenna transmission

4 General

4.1 Relationship between minimum requirements and test requirements

The present document is a Single-RAT specification for NR UE, covering RF characteristics and minimum performance requirements. Conformance to the present specification is demonstrated by fulfilling the test requirements specified in the conformance specification 3GPP TS 38.521-1 [4].

The Minimum Requirements given in this specification make no allowance for measurement uncertainty. The test specification TS 38.521-1 [4] defines test tolerances. These test tolerances are individually calculated for each test. The test tolerances are used to relax the minimum requirements in this specification to create test requirements. For some requirements, including regulatory requirements, the test tolerance is set to zero.

The measurement results returned by the test system are compared - without any modification - against the test requirements as defined by the shared risk principle.

The shared risk principle is defined in Recommendation ITUR M.1545 [5].

4.2 Applicability of minimum requirements

- a) In this specification the Minimum Requirements are specified as general requirements and additional requirements. Where the Requirement is specified as a general requirement, the requirement is mandated to be met in all scenarios
- b) For specific scenarios for which an additional requirement is specified, in addition to meeting the general requirement, the UE is mandated to meet the additional requirements.
- c) The spurious emissions power requirements are for the long-term average of the power. For the purpose of reducing measurement uncertainty it is acceptable to average the measured power over a period of time sufficient to reduce the uncertainty due to the statistical nature of the signal
- d) All the requirements for intra-band contiguous and non-contiguous CA apply under the assumption of the same slot format indicated by *TDD-UL-DL-ConfigurationCommon* and *TDD-UL-DL-ConfigurationDedicated* in the PCell and SCells for NR SA.

4.3 Specification suffix information

Unless stated otherwise the following suffixes are used for indicating at 2nd level clause, shown in Table 4.3-1.

Table 4.3-1: Definition of suffixes

Clause suffix	Variant
None	Single Carrier
A	Carrier Aggregation (CA)
B	Dual-Connectivity (DC)
C	Supplementary Uplink (SUL)
D	UL MIMO

A terminal which supports the above features needs to meet both the general requirements and the additional requirement applicable to the additional clause (suffix A, B, C and D) in clauses 5, 6 and 7. Where there is a difference in requirement between the general requirements and the additional clause requirements (suffix A, B, C and D) in clauses 5, 6 and 7, the tighter requirements are applicable unless stated otherwise in the additional clause.

A terminal which supports more than one feature in clauses 5, 6 and 7 shall meet all of the separate corresponding requirements.

For a terminal that supports SUL for the band combination specified in Table 5.2C-1, the current version of the specification assumes the terminal is configured with active transmission either on UL carrier or SUL carrier at any time in one serving cell and the UE requirements for single carrier shall apply for the active UL or SUL carrier accordingly. For a terminal that supports SUL, the current version of the specification assumes the terminal is not configured with UL MIMO on SUL carrier.

5 Operating bands and channel arrangement

5.1 General

The channel arrangements presented in this clause are based on the operating bands and channel bandwidths defined in the present release of specifications.

NOTE: Other operating bands and channel bandwidths may be considered in future releases.

Requirements throughout the RF specifications are in many cases defined separately for different frequency ranges (FR). The frequency ranges in which NR can operate according to this version of the specification are identified as described in Table 5.1-1.

Table 5.1-1: Definition of frequency ranges

Frequency range designation	Corresponding frequency range
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FR1	410 MHz – 7125 MHz
FR2	24250 MHz – 52600 MHz

The present specification covers FR1 operating bands.

5.2 Operating bands

NR is designed to operate in the FR1 operating bands defined in Table 5.2-1.

Table 5.2-1: NR operating bands in FR1

NR operating band	Uplink (UL) <i>operating band</i> BS receive / UE transmit $F_{UL_low} - F_{UL_high}$	Downlink (DL) <i>operating band</i> BS transmit / UE receive $F_{DL_low} - F_{DL_high}$	Duplex Mode
n1	1920 MHz – 1980 MHz	2110 MHz – 2170 MHz	FDD
n2	1850 MHz – 1910 MHz	1930 MHz – 1990 MHz	FDD
n3	1710 MHz – 1785 MHz	1805 MHz – 1880 MHz	FDD
n5	824 MHz – 849 MHz	869 MHz – 894 MHz	FDD
n7	2500 MHz – 2570 MHz	2620 MHz – 2690 MHz	FDD
n8	880 MHz – 915 MHz	925 MHz – 960 MHz	FDD
n12	699 MHz – 716 MHz	729 MHz – 746 MHz	FDD
n20	832 MHz – 862 MHz	791 MHz – 821 MHz	FDD
n25	1850 MHz – 1915 MHz	1930 MHz – 1995 MHz	FDD
n28	703 MHz – 748 MHz	758 MHz – 803 MHz	FDD
n34	2010 MHz – 2025 MHz	2010 MHz – 2025 MHz	TDD
n38	2570 MHz – 2620 MHz	2570 MHz – 2620 MHz	TDD
n39	1880 MHz – 1920 MHz	1880 MHz – 1920 MHz	TDD
n40	2300 MHz – 2400 MHz	2300 MHz – 2400 MHz	TDD
n41	2496 MHz – 2690 MHz	2496 MHz – 2690 MHz	TDD
n50	1432 MHz – 1517 MHz	1432 MHz – 1517 MHz	TDD ¹
n51	1427 MHz – 1432 MHz	1427 MHz – 1432 MHz	TDD
n66	1710 MHz – 1780 MHz	2110 MHz – 2200 MHz	FDD
n70	1695 MHz – 1710 MHz	1995 MHz – 2020 MHz	FDD
n71	663 MHz – 698 MHz	617 MHz – 652 MHz	FDD
n74	1427 MHz – 1470 MHz	1475 MHz – 1518 MHz	FDD
n75	N/A	1432 MHz – 1517 MHz	SDL ^{2,3}
n76	N/A	1427 MHz – 1432 MHz	SDL ³

n77	3300 MHz – 4200 MHz	3300 MHz – 4200 MHz	TDD
n78	3300 MHz – 3800 MHz	3300 MHz – 3800 MHz	TDD
n79	4400 MHz – 5000 MHz	4400 MHz – 5000 MHz	TDD
n80	1710 MHz – 1785 MHz	N/A	SUL
n81	880 MHz – 915 MHz	N/A	SUL
n82	832 MHz – 862 MHz	N/A	SUL
n83	703 MHz – 748 MHz	N/A	SUL
n84	1920 MHz – 1980 MHz	N/A	SUL
n86	1710 MHz – 1780 MHz	N/A	SUL
NOTE 1: UE that complies with the NR Band n50 minimum requirements in this specification shall also comply with the NR Band n51 minimum requirements.			
NOTE 2: UE that complies with the NR Band n75 minimum requirements in this specification shall also comply with the NR Band n76 minimum requirements.			
NOTE 3: For SDL bands, downlink configuration for RRM performance testing is same as FDD.			

5.2A Operating bands for CA

5.2A.1 Intra-band CA

NR intra-band contiguous carrier aggregation is designed to operate in the operating bands defined in Table 5.2A.1-1, where all operating bands are within FR1.

Table 5.2A.1-1: Intra-band contiguous CA operating bands in FR1

NR CA Band	NR Band (Table 5.2-1)
CA_n41 ¹	n41
CA_n77 ¹	n77
CA_n78 ¹	n78
CA_n79 ¹	n79
NOTE 1: The minimum requirements only apply for nonsimultaneous Tx/Rx between all carriers.	

5.2A.2 Inter-band CA

NR inter-band carrier aggregation is designed to operate in the operating bands defined in Table 5.2A.2.1-1, where all operating bands are within FR1.

If the mandatory simultaneous Rx/Tx capability applies for a lower order band combination, when the applicable lower order band combination is a band pair in a higher order band combination, the mandatory simultaneous Rx/Tx capability also applies for the band pair in the higher order band combination.

Table 5.2A.2-1: Void

5.2A.2.1 Inter-band CA (two bands)

Table 5.2A.2.1-1: Inter-band CA operating bands involving FR1 (two bands)

NR CA Band	NR Band (Table 5.2-1)
CA_n3-n77 ¹	n3, n77
CA_n3-n78 ¹	n3, n78
CA_n3-n79 ¹	n3, n79
CA_n8-n75 ¹	n8, n75
CA_n8-n78 ¹	n8, n78
CA_n8-n79 ¹	n8, n79
CA_n28-n75 ²	n28, n75
CA_n28-n78 ¹	n28, n78
CA_n41-n78	n41, n78
CA_n75-n78 ¹	n75, n78
CA_n76-n78 ¹	n76, n78
CA_n77-n79 ⁴	n77, n79
CA_n78-n79 ³	n78, n79
NOTE 1: Applicable for UE supporting inter-band carrier aggregation with mandatory simultaneous Rx/Tx capability. NOTE 2: The frequency range in band n28 is restricted for this band combination to 703-733 MHz for the UL and 758-788 MHz for the DL. NOTE 3: For UEs supporting band n77, the minimum requirements apply only when there is non-simultaneous Rx/Tx operation between n78-n79 NR carriers. This restriction applies also for these	

NOTE 4:	carriers when applicable NR CA configuration is part of a higher order configuration. The minimum requirements apply only when there is non-simultaneous Rx/Tx operation between n77-n79 NR carriers. This restriction applies also for these carriers when applicable NR CA configuration is part of a higher order configuration.
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5.2B Void

5.2C Operating band combination for SUL

NR operation is designed to operate in the operating band combination defined in Table 5.2C-1, where all operating bands are within FR1.

If the mandatory simultaneous Rx/Tx capability applies for a lower order band combination, when the applicable lower order band combination is a band pair in a higher order band combination, the mandatory simultaneous Rx/Tx capability also applies for the band pair in the higher order band combination.

Table 5.2C-1: Operating band combination for SUL in FR1

NR Band combination for SUL	NR Band (Table 5.2-1)
SUL_n78-n80 ²	n78, n80
SUL_n78-n81 ²	n78, n81
SUL_n78-n82 ²	n78, n82
SUL_n78-n83 ²	n78, n83
SUL_n78-n84 ²	n78, n84
SUL_n78-n86 ²	n78, n86
SUL_n79-n80 ²	n79, n80
SUL_n79-n81 ²	n79, n81
NOTE 1: If a UE is configured with both NR UL and NR SUL carriers in a cell, the switching time between NR UL carrier and NR SUL carrier is 0 us.	
NOTE 2: For UE supporting SUL band combination simultaneous Rx/Tx capability is mandatory.	
NOTE 3: For UE supporting SUL band combination, UL	

MIMO is not configured on SUL carrier.

5.3 UE channel bandwidth

5.3.1 General

The UE channel bandwidth supports a single NR RF carrier in the uplink or downlink at the UE. From a BS perspective, different UE channel bandwidths may be supported within the same spectrum for transmitting to and receiving from UEs connected to the BS. Transmission of multiple carriers to the same UE (CA) or multiple carriers to different UEs within the BS channel bandwidth can be supported.

From a UE perspective, the UE is configured with one or more BWP / carriers, each with its own UE channel bandwidth. The UE does not need to be aware of the BS channel bandwidth or how the BS allocates bandwidth to different UEs. The placement of the UE channel bandwidth for each UE carrier is flexible but can only be completely within the BS channel bandwidth.

The relationship between the channel bandwidth, the guardband and the maximum transmission bandwidth configuration is shown in Figure 5.3.1-1.

Figure 5.3.1-1: Definition of the channel bandwidth and the maximum transmission bandwidth configuration for one NR channel

5.3.2 Maximum transmission bandwidth configuration

The maximum transmission bandwidth configuration N_{RB} for each UE channel bandwidth and subcarrier spacing is specified in Table 5.3.2-1.

Table 5.3.2-1: Maximum transmission bandwidth configuration N_{RB}

	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
SCS (kHz)	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}

15	25	52	79	106	133	160	216	270	N/A	N/A	N/A	N/A
30	11	24	38	51	65	78	106	133	162	217	245	273
60	N/A	11	18	24	31	38	51	65	79	107	121	135

5.3.3 Minimum guardband and transmission bandwidth configuration

The minimum guardband for each UE channel bandwidth and SCS is specified in Table 5.3.3-1,

Table 5.3.3-1: Minimum guardband for each UE channel bandwidth and SCS (kHz)

SCS (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
15	242.5	312.5	382.5	452.5	522.5	592.5	552.5	692.5	N/A	N/A	N/A	N/A
30	505	665	645	805	785	945	905	1045	825	925	885	845
60	N/A	1010	990	1330	1310	1290	1610	1570	1530	1450	1410	1370

NOTE: The minimum guardbands have been calculated using the following equation: $GB_{channel} = (BW_{Channel} \times 1000 \text{ (kHz)} - N_{RB} \times SCS \times 12) / 2 - SCS/2$, where N_{RB} are from Table 5.3.2-1 and $GB_{channel}$ expressed in kHz.

Figure 5.3.3-1: Void

The number of RBs configured in any channel bandwidth shall ensure that the minimum guardband specified in this clause is met.

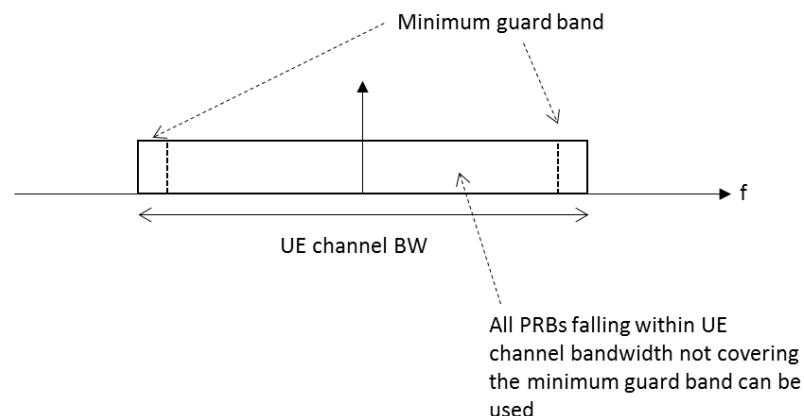


Figure 5.3.3-2: UE PRB utilization

In the case that multiple numerologies are multiplexed in the same symbol due to BS transmission of SSB, the minimum guardband on each side of the carrier is the guardband applied at the configured channel bandwidth for the numerology that is received immediately adjacent to the guard.

If multiple numerologies are multiplexed in the same symbol and the UE channel bandwidth is >50 MHz, the minimum guardband applied adjacent to 15 kHz SCS shall be the same as the minimum guardband defined for 30 kHz SCS for the same UE channel bandwidth.

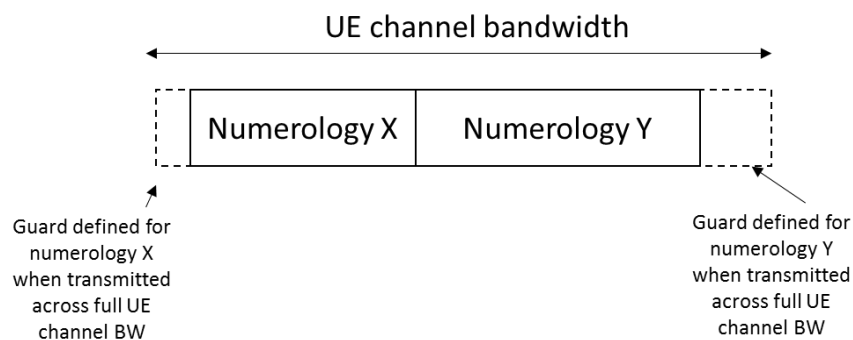


Figure 5.3.3-3 Guard band definition when transmitting multiple numerologies

NOTE: Figure 5.3.3-3 is not intended to imply the size of any guard between the two numerologies. Inter-numerology guard band within the carrier is implementation dependent.

5.3.4 RB alignment

For each numerology, its common resource blocks are specified in Clause 4.4.4.3 in TS 38.211 [6], and the starting point of its transmission bandwidth configuration on the common resource block grid for a given channel bandwidth is indicated by an offset to "Reference point A" in the unit of the numerology. The *UE transmission bandwidth configuration* is indicated by the higher layer parameter *carrierBandwidth* [7] and will fulfil the minimum UE guardband requirement specified in Clause 5.3.3.

5.3.5 UE channel bandwidth per operating band

The requirements in this specification apply to the combination of channel bandwidths, SCS and operating bands shown in Table 5.3.5-1. The transmission bandwidth configuration in Table 5.3.2-1 shall be supported for each of the specified channel bandwidths. The channel bandwidths are specified for both the TX and RX path.

Table 5.3.5-1 Channel bandwidths for each NR band

NR Band	NR band / SCS / UE Channel bandwidth												
	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 ⁴ MHz	100 MHz
n1	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n2	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n3	15	Yes	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes	Yes						
n5	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								

	60												
n7	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n8	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60												
n12	15	Yes	Yes	Yes									
	30		Yes	Yes									
	60												
n20	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60												
n25	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n28	15	Yes	Yes	Yes	Yes ⁵								
	30		Yes	Yes	Yes ⁵								
	60												
n34	15	Yes	Yes	Yes									
	30		Yes	Yes									
	60		Yes	Yes									
n38	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n39	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes	Yes					
	60		Yes	Yes	Yes	Yes	Yes	Yes					
n40	15	Yes ⁹	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
	30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
	60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
n41	15		Yes	Yes	Yes			Yes	Yes				
	30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes
	60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes

n50	15	Yes ⁹	Yes	Yes	Yes			Yes	Yes				
	30		Yes	Yes	Yes			Yes	Yes	Yes	Yes ³		
	60		Yes	Yes	Yes			Yes	Yes	Yes	Yes ³		
n51	15	Yes											
	30												
	60												
n66	15	Yes	Yes	Yes	Yes			Yes					
	30		Yes	Yes	Yes			Yes					
	60		Yes	Yes	Yes			Yes					
n70	15	Yes	Yes	Yes	Yes ³	Yes ³							
	30		Yes	Yes	Yes ³	Yes ³							
	60		Yes	Yes	Yes ³	Yes ³							
n71	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60												
n74	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n75	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n76	15	Yes											
	30												
	60												
n77	15		Yes	Yes	Yes			Yes	Yes				
	30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes
	60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes
n78	15		Yes	Yes	Yes			Yes	Yes				
	30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes
	60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes
n79	15							Yes	Yes				
	30							Yes	Yes	Yes	Yes		Yes

	60							Yes	Yes	Yes	Yes		Yes
n80	15	Yes	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes	Yes						
n81	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60												
n82	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60												
n83	15	Yes	Yes	Yes	Yes ⁵								
	30		Yes	Yes	Yes ⁵								
	60												
n84	15	Yes	Yes	Yes	Yes								
	30		Yes	Yes	Yes								
	60		Yes	Yes	Yes								
n86	15	Yes	Yes	Yes	Yes			Yes					
	30		Yes	Yes	Yes			Yes					
	60		Yes	Yes	Yes			Yes					
NOTE 1: void NOTE 2: void NOTE 3: This UE channel bandwidth is applicable only to downlink. NOTE 4: This UE channel bandwidth is optional in this release of the specification. NOTE 5: For the 20 MHz bandwidth, the minimum requirements are specified for NR UL carrier frequencies confined to either 713-723 MHz or 728-738 MHz. NOTE 6: Void. NOTE 7: Void. NOTE 8: Void. NOTE 9: For this bandwidth, the minimum requirements are restricted to operation when carrier is configured as an SCell part of DC or CA configuration.													

5.3.6 Asymmetric channel bandwidths

The UE channel bandwidth can be asymmetric in downlink and uplink. In asymmetric channel bandwidth operation, the narrower carrier shall be confined within the frequency range of the wider channel bandwidth.

In FDD, the confinement is defined as a deviation to the default Tx-Rx carrier center frequency separation (defined in table 5.4.4-1) as following:

$$\Delta F_{\text{TX-RX}} = | (BW_{\text{DL}} - BW_{\text{UL}}) / 2 |$$

The operating bands and supported asymmetric channel bandwidth combinations are defined in table 5.3.6-1.

Table 5.3.6-1: FDD asymmetric UL and DL channel bandwidth combinations

NR Band	Channel bandwidths for UL (MHz)	Channel bandwidths for DL (MHz)	Asymmetric channel bandwidth combination set
n66	5, 10	20, 40	0
	20	40	
n70	5, 10	15	0
	5, 10, 15	20, 25	
n71	5	10	0
	10	15	
	15	20	

In TDD, the operating bands and supported asymmetric channel bandwidth combinations are defined in table 5.3.6-2.

Table 5.3.6-2: TDD asymmetric UL and DL channel bandwidth combinations

NR Band	Channel bandwidths for UL (MHz)	Channel bandwidths for DL (MHz)	Asymmetric channel bandwidth combination set
n50	60	80	0
NOTE 1: Both centre frequency and BWP-ID shall match between DL and UL carriers as defined in TS 38.331 [7] cl. 6.3.2 and TS 38.213 [8] section 12. NOTE 2: In a case a UE is configured with a full width of BWP within both UL/ DL channels, the centre frequency of UL/ DL channels shall be same. NOTE 3: A position of Point A is common between UL and DL carriers as defined in TS 38.331 [7] cl. 6.3.2. NOTE 4: The maximum transmission bandwidth configuration N_{RB} for 60 MHz UL channel BW is reduced from the value defined in table 5.3.2-1 to 161 for 30kHz SCS for Asymmetric channel bandwidth combination set 0 for n50.			

Offset, high	

Resource block

where $N_{RB,low}$ and $N_{RB,high}$ are the transmission bandwidth configurations according to Table 5.3.2-1 for the lowest and highest assigned component carrier within a sub-block, respectively. SCS_{low} and SCS_{high} are the sub-carrier spacing for the lowest and highest assigned component carrier within a sub-block, respectively. SCS_{low} , SCS_{high} , $N_{RB,low}$, $N_{RB,high}$, and $BW_{GB,Channel(k)}$ use the largest μ value among the subcarrier spacing configurations supported in the operating band for both of the channel bandwidths according to Table 5.3.5-1 and $BW_{GB,Channel(k)}$ is the minimum guard band for carrier k according to Table 5.3.3-1 for the said μ value. In case there is no common μ value for both of the channel bandwidths, $\mu=1$ is used for SCS_{low} , SCS_{high} , $N_{RB,low}$, $N_{RB,high}$, and $BW_{GB,Channel(k)}$. The sub-block gap size between two consecutive sub-blocks W_{gap} is defined as

$$W_{gap} = F_{edge,block\ n+1,low} - F_{edge,block\ n,high} \text{ (MHz)}$$

5.3A.4 Void

5.3A.5 UE channel bandwidth per operating band for CA

The requirements for carrier aggregation in this specification are defined for carrier aggregation configurations. For intra-band contiguous carrier aggregation, a carrier aggregation configuration is a single operating band supporting a carrier aggregation bandwidth class with associated bandwidth combination sets specified in clause 5.5A.1. For each carrier aggregation configuration, requirements are specified for all aggregated channel bandwidths contained in a bandwidth combination set, a UE can indicate support of several bandwidth combination sets per carrier aggregation configuration. For intra-band non-contiguous carrier aggregation, a carrier aggregation configuration is a single operating band supporting two or more sub-blocks, each supporting a carrier aggregation bandwidth class. For inter-band carrier aggregation, a carrier aggregation configuration is a combination of operating bands, each supporting a carrier aggregation bandwidth class.

Table 5.3A.5-1: NR CA bandwidth classes

NR CA bandwidth class	Aggregated channel bandwidth	Number of contiguous CC	Fallback group
A	$BW_{Channel} \leq BW_{Channel,max}$	1	1, 2
B	$20 \text{ MHz} \leq BW_{Channel_CA} \leq 100 \text{ MHz}$	2	2
C	$100 \text{ MHz} < BW_{Channel_CA} \leq 2 \times BW_{Channel,max}$	2	1

D	$200 \text{ MHz} < BW_{\text{Channel_CA}} \leq 3 \times BW_{\text{Channel,max}}$	3	2
E	$300 \text{ MHz} < BW_{\text{Channel_CA}} \leq 4 \times BW_{\text{Channel,max}}$	4	
G	$100 \text{ MHz} < BW_{\text{Channel_CA}} \leq 150 \text{ MHz}$	3	
H	$150 \text{ MHz} < BW_{\text{Channel_CA}} \leq 200 \text{ MHz}$	4	
I	$200 \text{ MHz} < BW_{\text{Channel_CA}} \leq 250 \text{ MHz}$	5	
J	$250 \text{ MHz} < BW_{\text{Channel_CA}} \leq 300 \text{ MHz}$	6	
K	$300 \text{ MHz} < BW_{\text{Channel_CA}} \leq 350 \text{ MHz}$	7	
L	$350 \text{ MHz} < BW_{\text{Channel_CA}} \leq 400 \text{ MHz}$	8	
NOTE 1: $BW_{\text{Channel,max}}$ is maximum channel bandwidth supported among all bands in a release			
NOTE 2: It is mandatory for a UE to be able to fallback to lower order NR CA bandwidth class configuration within a fallback group. It is not mandatory for a UE to be able to fallback to lower order NR CA bandwidth class configuration that belong to a different fallback group			

5.4 Channel arrangement

5.4.1 Channel spacing

5.4.1.1 Channel spacing for adjacent NR carriers

The spacing between carriers will depend on the deployment scenario, the size of the frequency block available and the channel bandwidths. The nominal channel spacing between two adjacent NR carriers is defined as following:

- For NR operating bands with 100 kHz channel raster,

$$\text{Nominal Channel spacing} = (BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)})/2$$

- For NR operating bands with 15 kHz channel raster,

$$\text{Nominal Channel spacing} = (BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)})/2 + \{-5 \text{ kHz}, 0 \text{ kHz}, 5 \text{ kHz}\} \text{ for } \Delta F_{\text{Raster}} \text{ equals } 15 \text{ kHz}$$

$$\text{Nominal Channel spacing} = (BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)})/2 + \{-10 \text{ kHz}, 0 \text{ kHz}, 10 \text{ kHz}\} \text{ for } \Delta F_{\text{Raster}} \text{ equals } 30 \text{ kHz}$$

where $BW_{\text{Channel}(1)}$ and $BW_{\text{Channel}(2)}$ are the channel bandwidths of the two respective NR carriers. The channel spacing can be adjusted depending on the channel raster to optimize performance in a particular deployment scenario.

5.4.2 Channel raster

5.4.2.1 NR-ARFCN and channel raster

The global frequency channel raster defines a set of RF reference frequencies F_{REF} . The RF reference frequency is used in signalling to identify the position of RF channels, SS blocks and other elements.

The global frequency raster is defined for all frequencies from 0 to 100 GHz. The granularity of the global frequency raster is ΔF_{Global} .

RF reference frequencies are designated by an NR Absolute Radio Frequency Channel Number (NR-ARFCN) in the range (0... 2016666) on the global frequency raster. The relation between the NR-ARFCN and the RF reference frequency F_{REF} in MHz is given by the following equation, where $F_{\text{REF-Offs}}$ and $N_{\text{Ref-Offs}}$ are given in table 5.4.2.1-1 and N_{REF} is the NR-ARFCN.

$$F_{\text{REF}} = F_{\text{REF-Offs}} + \Delta F_{\text{Global}} (N_{\text{REF}} - N_{\text{REF-Offs}})$$

Table 5.4.2.1-1: NR-ARFCN parameters for the global frequency raster

Frequency range (MHz)	ΔF_{Global} (kHz)	$F_{\text{REF-Offs}}$ (MHz)	$N_{\text{REF-Offs}}$	Range of N_{REF}
0 – 3000	5	0	0	0 – 599999
3000 – 24250	15	3000	600000	600000 – 2016666

The channel raster defines a subset of RF reference frequencies that can be used to identify the RF channel position in the uplink and downlink. The RF reference frequency for an RF channel maps to a resource element on the carrier. For each operating band, a subset of frequencies from the global frequency raster are applicable for that band and forms a channel raster with a granularity ΔF_{Raster} , which may be equal to or larger than ΔF_{Global} .

For SUL bands, for the uplink of all FDD bands defined in Table 5.2-1, and for TDD band n34, n38, and n39

$$F_{\text{REF, shift}} = F_{\text{REF}} + \Delta_{\text{shift}}, \Delta_{\text{shift}} = 0 \text{ kHz or } 7.5 \text{ kHz.}$$

where Δ_{shift} is signalled by the network in higher layer parameter *frequencyShift7p5khz* [7]. For band n34, n38, n39 $F_{\text{REF, shift}}$ is only applicable to uplink transmissions using a 15 kHz SCS.

NOTE: A UE operating n34, n38, n39 should support application of $F_{\text{REF, shift}}$ for UL transmissions. A UE that does not support it will be unable to communicate with a network that signals $\Delta_{\text{shift}} = 7.5$ kHz.

The mapping between the channel raster and corresponding resource element is given in Clause 5.4.2.2. The applicable entries for each operating band are defined in Clause 5.4.2.3.

5.4.2.2 Channel raster to resource element mapping

The mapping between the RF reference frequency on the channel raster and the corresponding resource element is given in Table 5.4.2.2-1 and can be used to identify the RF channel position. The mapping depends on the total number of RBs that are allocated in the channel and applies to both UL and DL. The mapping must apply to at least one numerology supported by the UE.

Table 5.4.2.2-1: Channel raster to resource element mapping

	$N_{\text{RB}} \bmod 2 = 0$	$N_{\text{RB}} \bmod 2 = 1$
Resource element index	0	6
Physical resource block number		

, n_{PRB} , N_{RB} are as defined in TS 38.211[6].

5.4.2.3 Channel raster entries for each operating band

The RF channel positions on the channel raster in each NR operating band are given through the applicable NR-ARFCN in Table 5.4.2.31, using the channel raster to resource element mapping in clause 5.4.2.2.

For NR operating bands with 100 kHz channel raster, $\Delta F_{\text{Raster}} = 20 \times \Delta F_{\text{Global}}$. In this case every 20th NR-ARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in Table 5.4.2.31 is given as <20>.

For NR operating bands with 15 kHz channel raster below 3GHz, $\Delta F_{\text{Raster}} = I \times \Delta F_{\text{Global}}$, where $I \in \{3, 6\}$. Every I^{th} NRARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in Table 5.4.2.31 is given as <I>.

For NR operating bands with 15 kHz channel raster above 3GHz, $\Delta F_{\text{Raster}} = I \times \Delta F_{\text{Global}}$, where $I \in \{1,2\}$. Every I^{th} NRARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in table 5.4.2.3-1 is given as $\langle I \rangle$.

In frequency bands with two ΔF_{Raster} , the higher ΔF_{Raster} applies to channels using only the SCS that is equal to or larger than the higher ΔF_{Raster} and SSB SCS is equal to the higher ΔF_{Raster} .

Table 5.4.2.3-1: Applicable NR-ARFCN per operating band

NR operating band	ΔF_{Raster} (kHz)	Uplink Range of N_{REF} (First - $\langle \text{Step size} \rangle$ - Last)	Downlink Range of N_{REF} (First - $\langle \text{Step size} \rangle$ - Last)
n1	100	384000 - $\langle 20 \rangle$ - 396000	422000 - $\langle 20 \rangle$ - 434000
n2	100	370000 - $\langle 20 \rangle$ - 382000	386000 - $\langle 20 \rangle$ - 398000
n3	100	342000 - $\langle 20 \rangle$ - 357000	361000 - $\langle 20 \rangle$ - 376000
n5	100	164800 - $\langle 20 \rangle$ - 169800	173800 - $\langle 20 \rangle$ - 178800
n7	100	500000 - $\langle 20 \rangle$ - 514000	524000 - $\langle 20 \rangle$ - 538000
n8	100	176000 - $\langle 20 \rangle$ - 183000	185000 - $\langle 20 \rangle$ - 192000
n12	100	139800 - $\langle 20 \rangle$ - 143200	145800 - $\langle 20 \rangle$ - 149200
n20	100	166400 - $\langle 20 \rangle$ - 172400	158200 - $\langle 20 \rangle$ - 164200
n25	100	370000 - $\langle 20 \rangle$ - 383000	386000 - $\langle 20 \rangle$ - 399000
n28	100	140600 - $\langle 20 \rangle$ - 149600	151600 - $\langle 20 \rangle$ - 160600
n34	100	402000 - $\langle 20 \rangle$ - 405000	402000 - $\langle 20 \rangle$ - 405000
n38	100	514000 - $\langle 20 \rangle$ - 524000	514000 - $\langle 20 \rangle$ - 524000
n39	100	376000 - $\langle 20 \rangle$ - 384000	376000 - $\langle 20 \rangle$ - 384000
n40	100	460000 - $\langle 20 \rangle$ - 480000	460000 - $\langle 20 \rangle$ - 480000
n41	15	499200 - $\langle 3 \rangle$ - 537999	499200 - $\langle 3 \rangle$ - 537999
	30	499200 - $\langle 6 \rangle$ - 537996	499200 - $\langle 6 \rangle$ - 537996
n50	100	286400 - $\langle 20 \rangle$ - 303400	286400 - $\langle 20 \rangle$ - 303400
n51	100	285400 - $\langle 20 \rangle$ - 286400	285400 - $\langle 20 \rangle$ - 286400
n66	100	342000 - $\langle 20 \rangle$ - 356000	422000 - $\langle 20 \rangle$ - 440000
n70	100	339000 - $\langle 20 \rangle$ - 342000	399000 - $\langle 20 \rangle$ - 404000
n71	100	132600 - $\langle 20 \rangle$ - 139600	123400 - $\langle 20 \rangle$ - 130400
n74	100	285400 - $\langle 20 \rangle$ - 294000	295000 - $\langle 20 \rangle$ - 303600

n75	100	N/A	286400 - <20> - 303400
n76	100	N/A	285400 - <20> - 286400
n77	15	620000 - <1> - 680000	620000 - <1> - 680000
	30	620000 - <2> - 680000	620000 - <2> - 680000
n78	15	620000 - <1> - 653333	620000 - <1> - 653333
	30	620000 - <2> - 653332	620000 - <2> - 653332
n79	15	693334 - <1> - 733333	693334 - <1> - 733333
	30	693334 - <2> - 733332	693334 - <2> - 733332
n80	100	342000 - <20> - 357000	N/A
n81	100	176000 - <20> - 183000	N/A
n82	100	166400 - <20> - 172400	N/A
n83	100	140600 - <20> - 149600	N/A
n84	100	384000 - <20> - 396000	N/A
n86	100	342000 - <20> - 356000	N/A

5.4.3 Synchronization raster

5.4.3.1 Synchronization raster and numbering

The synchronization raster indicates the frequency positions of the synchronization block that can be used by the UE for system acquisition when explicit signalling of the synchronization block position is not present.

A global synchronization raster is defined for all frequencies. The frequency position of the SS block is defined as SS_{REF} with corresponding number GSCN. The parameters defining the SS_{REF} and GSCN for all the frequency ranges are in Table 5.4.3.1-1. The resource element corresponding to the SS block reference frequency SS_{REF} is given in clause 5.4.3.2. The synchronization raster and the subcarrier spacing of the synchronization block is defined separately for each band.

Table 5.4.3.1-1: GSCN parameters for the global frequency raster

Frequency range	SS Block frequency position SS_{REF}	GSCN	Range of GSCN
0 – 3000 MHz	$N * 1200\text{kHz} + M * 50\text{ kHz}$, $N=1:2499$, $M \in \{1,3,5\}$ (Note 1)	$3N + (M-3)/2$	2 – 7498
3000 – 24250 MHz	$3000\text{ MHz} + N * 1.44\text{ MHz}$ $N = 0:14756$	$7499 + N$	7499 – 22255

NOTE 1: The default value for operating bands with SCS spaced channel raster is M=3.

5.4.3.2 Synchronization raster to synchronization block resource element mapping

The mapping between the synchronization raster and the corresponding resource element of the SS block is given in Table 5.4.3.2-1.

Table 5.4.3.2-1: Synchronization raster to SS block resource element mapping

Resource element index	120
------------------------	-----

is the subcarrier number of SS/PBCH block defined in TS 38.211 clause 7.4.3.1 [6].

5.4.3.3 Synchronization raster entries for each operating band

The synchronization raster for each band is give in Table 5.4.3.3-1. The distance between applicable GSCN entries is given by the <Step size> indicated in Table 5.4.3.3-1.

Table 5.4.3.3-1: Applicable SS raster entries per operating band

NR operating band	SS Block SCS	SS Block pattern ¹	Range of GSCN (First – <Step size> – Last)
n1	15 kHz	Case A	5279 – <1> – 5419
n2	15 kHz	Case A	4829 – <1> – 4969
n3	15 kHz	Case A	4517 – <1> – 4693
n5	15 kHz	Case A	2177 – <1> – 2230
	30 kHz	Case B	2183 – <1> – 2224
n7	15 kHz	Case A	6554 – <1> – 6718
n8	15 kHz	Case A	2318 – <1> – 2395
n12	15 kHz	Case A	1828 – <1> – 1858
n20	15 kHz	Case A	1982 – <1> – 2047
n25	15 kHz	Case A	4829 – <1> – 4981
n28	15 kHz	Case A	1901 – <1> – 2002
n34	15 kHz	Case A	NOTE 3
	30 kHz	Case C	5036 – <1> – 5050
n38	15 kHz	Case A	NOTE 2

	30 kHz	Case C	6437 – <1> – 6538
n39	15 kHz	Case A	NOTE 4
	30 kHz	Case C	4712 – <1> – 4789
n40	30 kHz	Case C	5762 – <1> – 5989
n41	15 kHz	Case A	6246 – <3> – 6717
	30 kHz	Case C	6252 – <3> – 6714
n50	30 kHz	Case C	3590 – <1> – 3781
n51	15 kHz	Case A	3572 – <1> – 3574
n66	15 kHz	Case A	5279 – <1> – 5494
	30 kHz	Case B	5285 – <1> – 5488
n70	15 kHz	Case A	4993 – <1> – 5044
n71	15 kHz	Case A	1547 – <1> – 1624
n74	15 kHz	Case A	3692 – <1> – 3790
n75	15 kHz	Case A	3584 – <1> – 3787
n76	15 kHz	Case A	3572 – <1> – 3574
n77	30 kHz	Case C	7711 – <1> – 8329
n78	30 kHz	Case C	7711 – <1> – 8051
n79	30 kHz	Case C	8480 – <16> – 8880
NOTE 1: SS Block pattern is defined in clause 4.1 in TS 38.213 [8] NOTE 2: The applicable SS raster entries are GSCN = {6432, 6443, 6457, 6468, 6479, 6493, 6507, 6518, 6532, 6543} NOTE 3: The applicable SS raster entries are GSCN = {5032, 5043, 5054} NOTE 4: The applicable SS raster entries are GSCN = {4707, 4715, 4718, 4729, 4732, 4743, 4747, 4754, 4761, 4768, 4772, 4782, 4786, 4793}			

5.4.4 TX–RX frequency separation

The default TX channel (carrier centre frequency) to RX channel (carrier centre frequency) separation for operating bands is specified in Table 5.4.4-1.

Table 5.4.4-1: Default UE TX-RX frequency separation

NR Operating Band	TX – RX carrier centre frequency separation
n1	190 MHz
n2	80 MHz
n3	95 MHz
n5	45 MHz

n7	120 MHz
n8	45 MHz
n12	30 MHz
n20	-41 MHz
n25	80 MHz
n28	55 MHz
n66	400 MHz
n70	300 MHz
n71	-46 MHz
n74	48 MHz
NOTE 1: Void	

5.4A Channel arrangement for CA

5.4A.1 Channel spacing for CA

For intra-band contiguous carrier aggregation with two or more component carriers, the nominal channel spacing between two adjacent NR component carriers is defined as the following unless stated otherwise:
For NR operating bands with a 100 kHz channel raster:

while for NR operating bands without a 100 kHz channel raster:

with

$$n = \mu_0$$

where $BW_{Channel(1)}$ and $BW_{Channel(2)}$ are the channel bandwidths of the two respective NR component carriers according to Table 5.3.2-1 with values in MHz, μ_0 is the largest μ value among the subcarrier spacing configurations supported in the operating band for both of the channel bandwidths according to Table 5.3.5-1 and $GB_{Channel(i)}$ is the minimum guard band for channel bandwidth i according to Table 5.3.3-1 for the said μ value with μ as defined in TS 38.211. In case there is no common μ value for both of the channel bandwidths, $\mu_0=1$ is selected and $GB_{Channel(i)}$ is the minimum guard band for channel bandwidth i according to Table 5.3.3-1 for $\mu=1$ with μ as defined in TS 38.211.

The channel spacing for intra-band contiguous carrier aggregation can be adjusted to any multiple of least common multiple of channel raster and sub-carrier spacing less than the nominal channel spacing to optimize performance in a particular deployment scenario.

For intra-band non-contiguous carrier aggregation, the channel spacing between two NR component carriers in different sub-blocks shall be larger than the nominal channel spacing defined in this clause.

5.4A.2 Channel raster for CA

For inter-band carrier aggregation, the channel raster requirements in clause 5.4.2 apply for each operating band.

5.4A.3 Synchronization raster for CA

For inter-band carrier aggregation, the synchronization raster requirements in clause 5.4.3 apply for each operating band.

5.4A.4 Tx-Rx frequency separation for CA

For inter-band carrier aggregation, the Tx-Rx frequency separation requirements in clause 5.4.4 apply for each operating band.

5.5 Void

5.5A Configurations for CA

5.5A.1 Configurations for intra-band contiguous CA

Table 5.5A.1-1: NR CA configurations and bandwidth combination sets defined for intra-band contiguous CA

NR CA configuration / Bandwidth combination set								
NR CA configuration	Uplink CA configurations	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Maximum aggregated bandwidth (MHz)	Bandwidth combination set
CA_n41C	-	40	80, 100				180	0
		50, 60, 80	60, 80, 100					
CA_n77C CA_n78C CA_n79C	-	50	60, 80, 100				200	0
		60	60, 80, 100					

		80	80, 100					
		100	100					
NOTE: Unless otherwise stated, minimum requirements are applicable irrespective of the order of the component carriers.								

Table 5.5A.1-2: Void

5.5A.3 Configurations for inter-band CA

5.5A.3.1 Configurations for inter-band CA (two bands)

NR CA configuration	Uplink CA configuration	NR Band	SCS (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz	Bandwidth combination set
CA_n3A-n77A	-	n3	15	Yes	Yes	Yes	Yes	Yes	Yes							0
			30		Yes	Yes	Yes	Yes	Yes							
			60		Yes	Yes	Yes	Yes	Yes							
		n77	15		Yes	Yes	Yes			Yes	Yes					
			30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
CA_n3A-n78A	CA_n3A-n78A	n3	15	Yes	Yes	Yes	Yes	Yes	Yes							0
			30		Yes	Yes	Yes	Yes	Yes							
			60		Yes	Yes	Yes	Yes	Yes							
		n78	15		Yes	Yes	Yes			Yes	Yes					
			30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
CA_n3A-n79A	-	n3	15	Yes	Yes	Yes	Yes	Yes	Yes							0
			30		Yes	Yes	Yes	Yes	Yes							
			60		Yes	Yes	Yes	Yes	Yes							
		n79	15		Yes	Yes	Yes			Yes	Yes					
			30		Yes	Yes	Yes			Yes	Yes	Yes	Yes		Yes	
			60		Yes	Yes	Yes			Yes	Yes	Yes	Yes		Yes	
CA_n8A-n75A	-	n8	15	Yes	Yes	Yes	Yes									0
			30		Yes	Yes	Yes									
			60													
		n75	15	Yes	Yes	Yes	Yes									
			30		Yes	Yes	Yes									
			60		Yes	Yes	Yes									

		60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
	n80	15	Yes	Yes	Yes	Yes	Yes	Yes							
SUL_n78A-n81A	n78	15		Yes	Yes	Yes			Yes	Yes					0
		30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
		60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
	n81	15	Yes	Yes	Yes	Yes									
SUL_n78A-n82A	n78	15		Yes	Yes	Yes			Yes	Yes					0
		30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
		60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
	n82	15	Yes	Yes	Yes	Yes									
SUL_n78A-n83A	n78	15		Yes	Yes	Yes			Yes	Yes					0
		30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
		60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
	n83	15	Yes	Yes	Yes	Yes									
SUL_n78A-n84A	n78	15		Yes	Yes	Yes			Yes	Yes					0
		30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
		60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
	n84	15	Yes	Yes	Yes	Yes									
SUL_n78A-n86A	n78	15		Yes	Yes	Yes			Yes	Yes					0
		30		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
		60		Yes	Yes	Yes			Yes	Yes	Yes	Yes	Yes	Yes	
	n86	15	Yes	Yes	Yes	Yes									
SUL_n79A-n80A	n79	15							Yes	Yes					0
		30							Yes	Yes	Yes	Yes		Yes	
		60							Yes	Yes	Yes	Yes		Yes	
	n80	15	Yes	Yes	Yes	Yes	Yes	Yes							
SUL_n79A-n81A	n79	15							Yes	Yes					0
		30							Yes	Yes	Yes	Yes		Yes	
		60							Yes	Yes	Yes	Yes		Yes	
	n81	15	Yes	Yes	Yes	Yes									

